

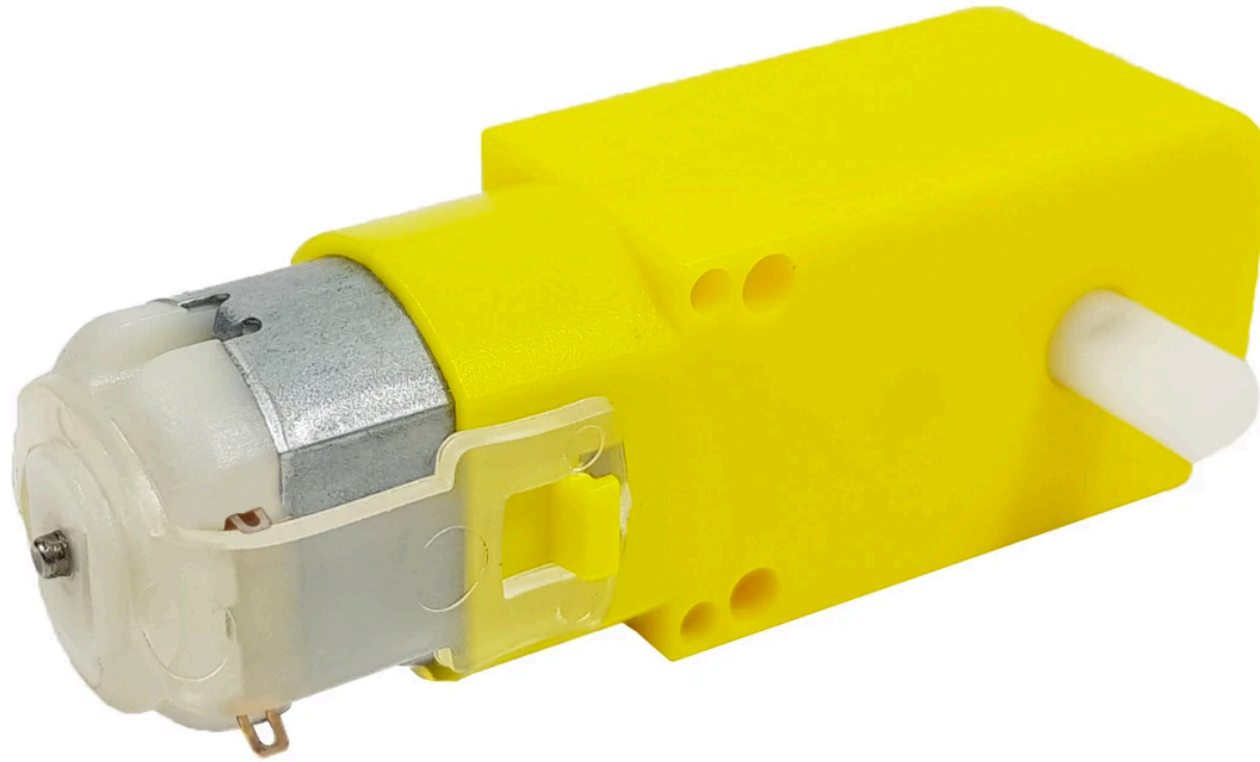
EG1311: Design & Make

Motor Physics

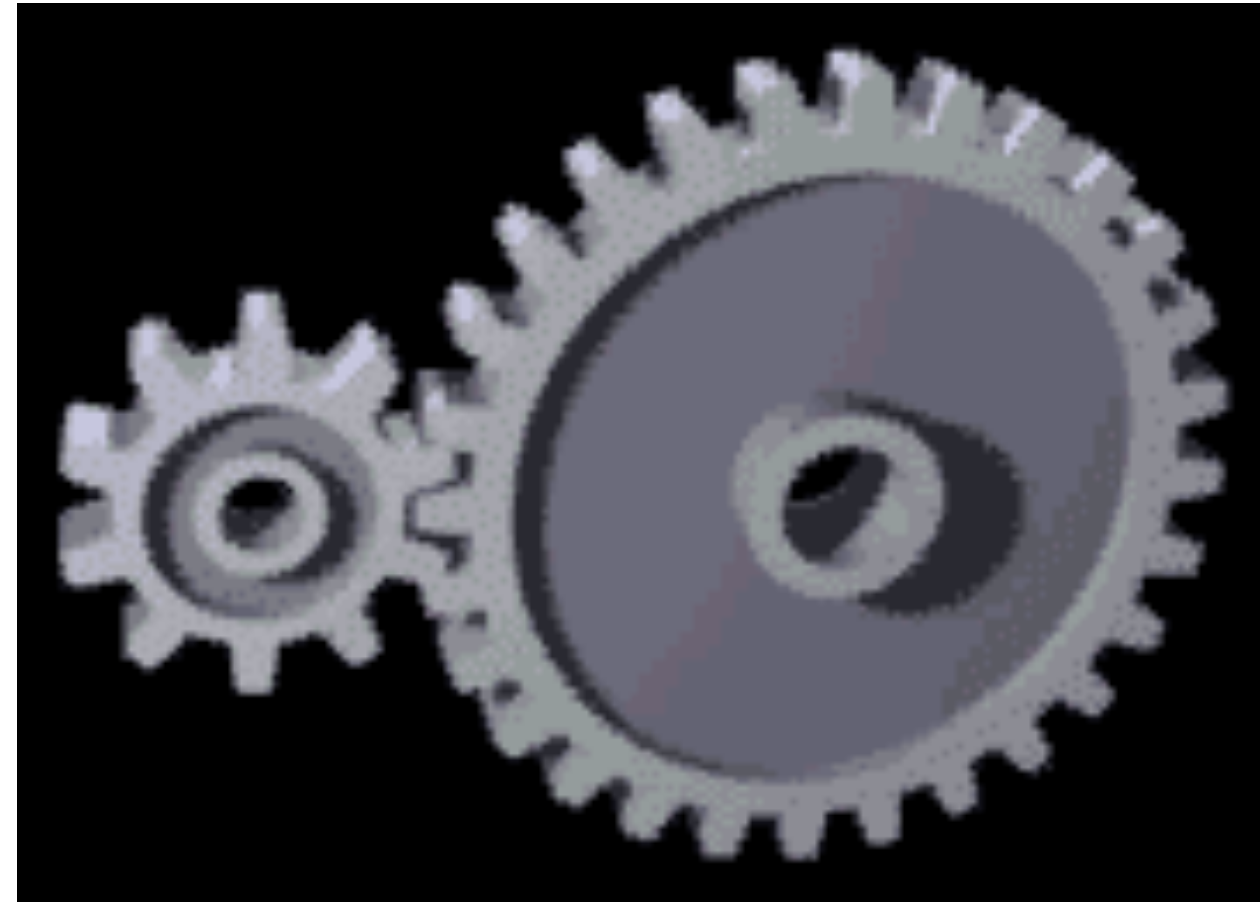
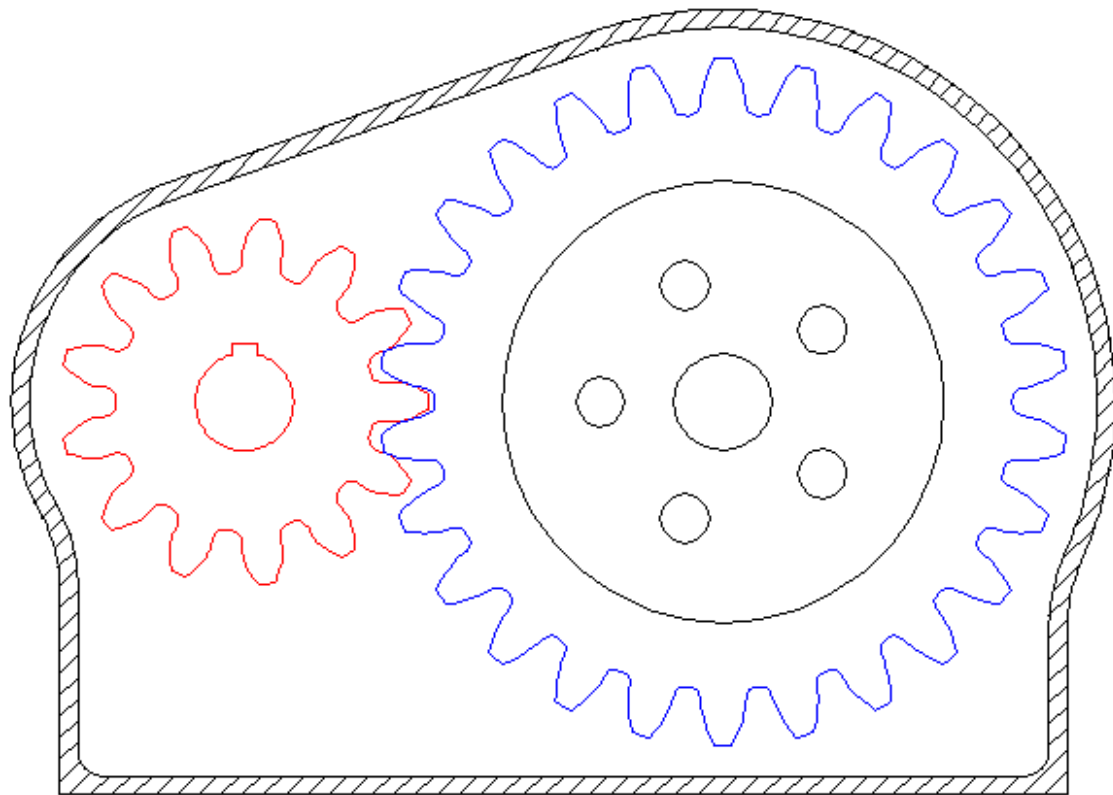
Dr. Jason S. Ku

DC Motor

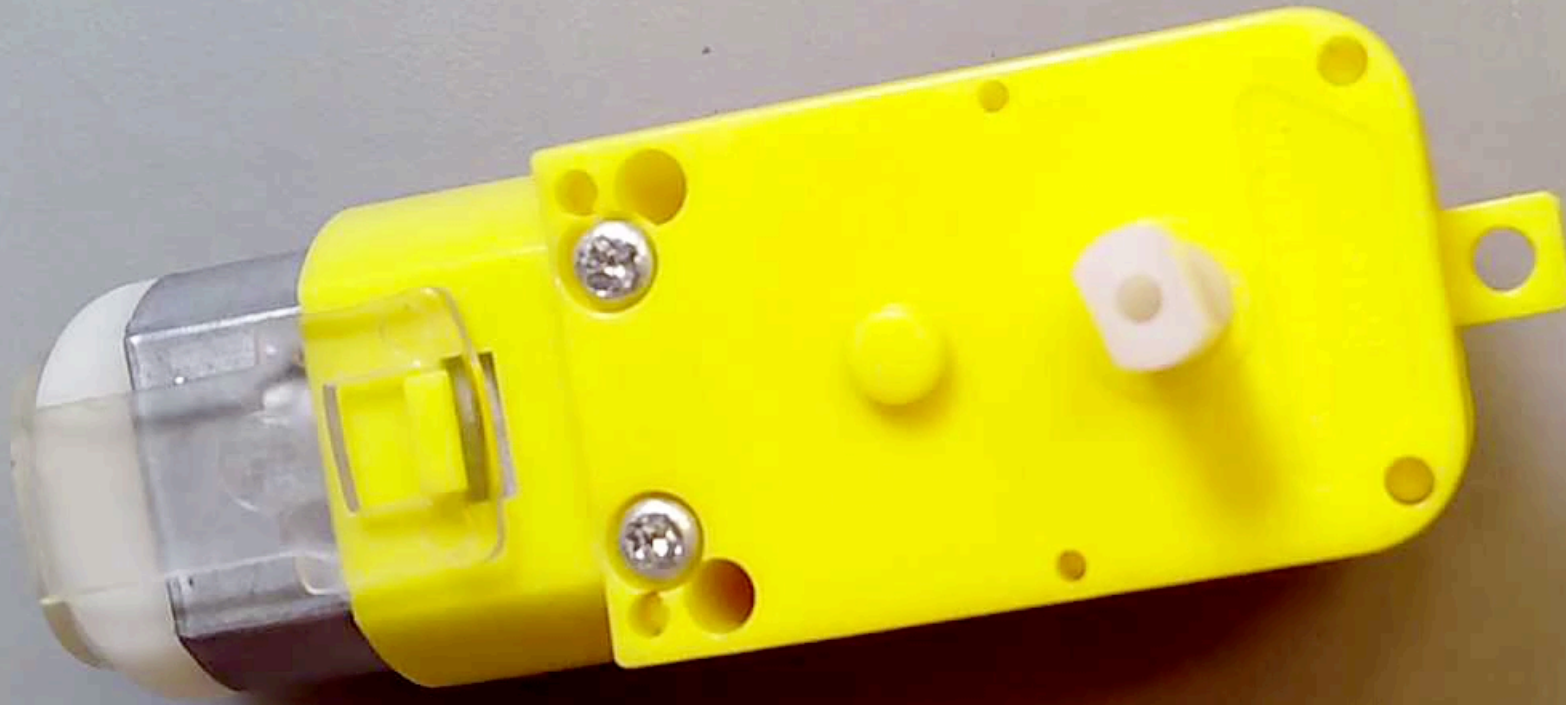
Motor



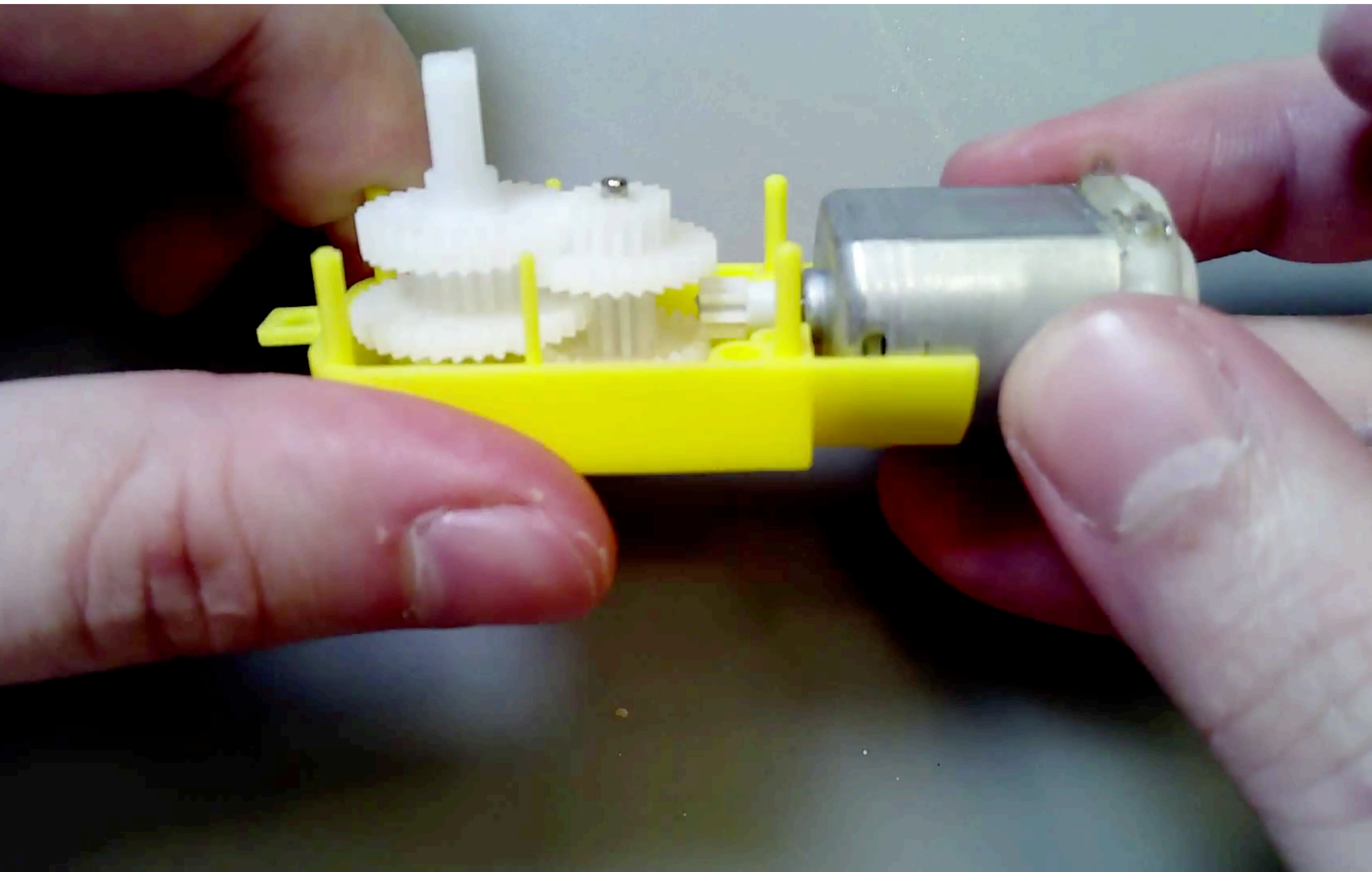
Gearbox



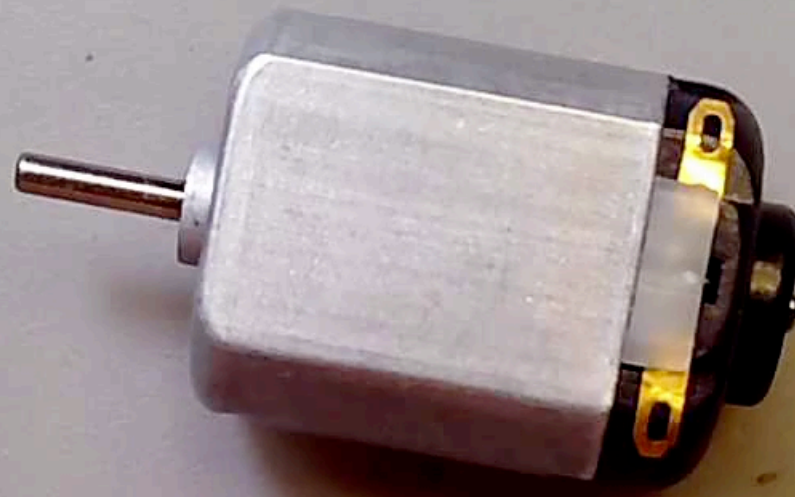
DC Motor



DC Motor



DC Motor

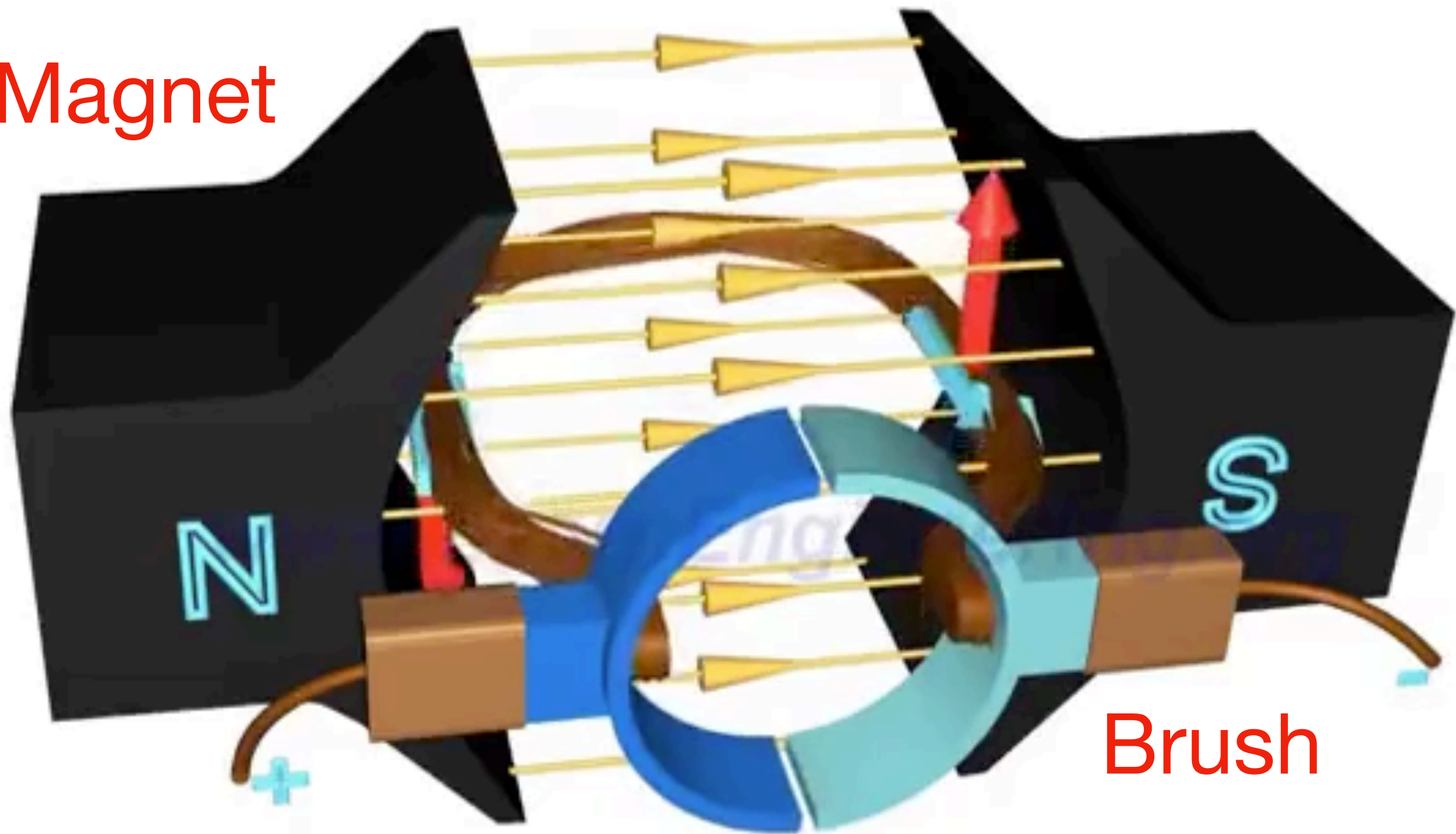


DC Motor



DC Motor

Magnet



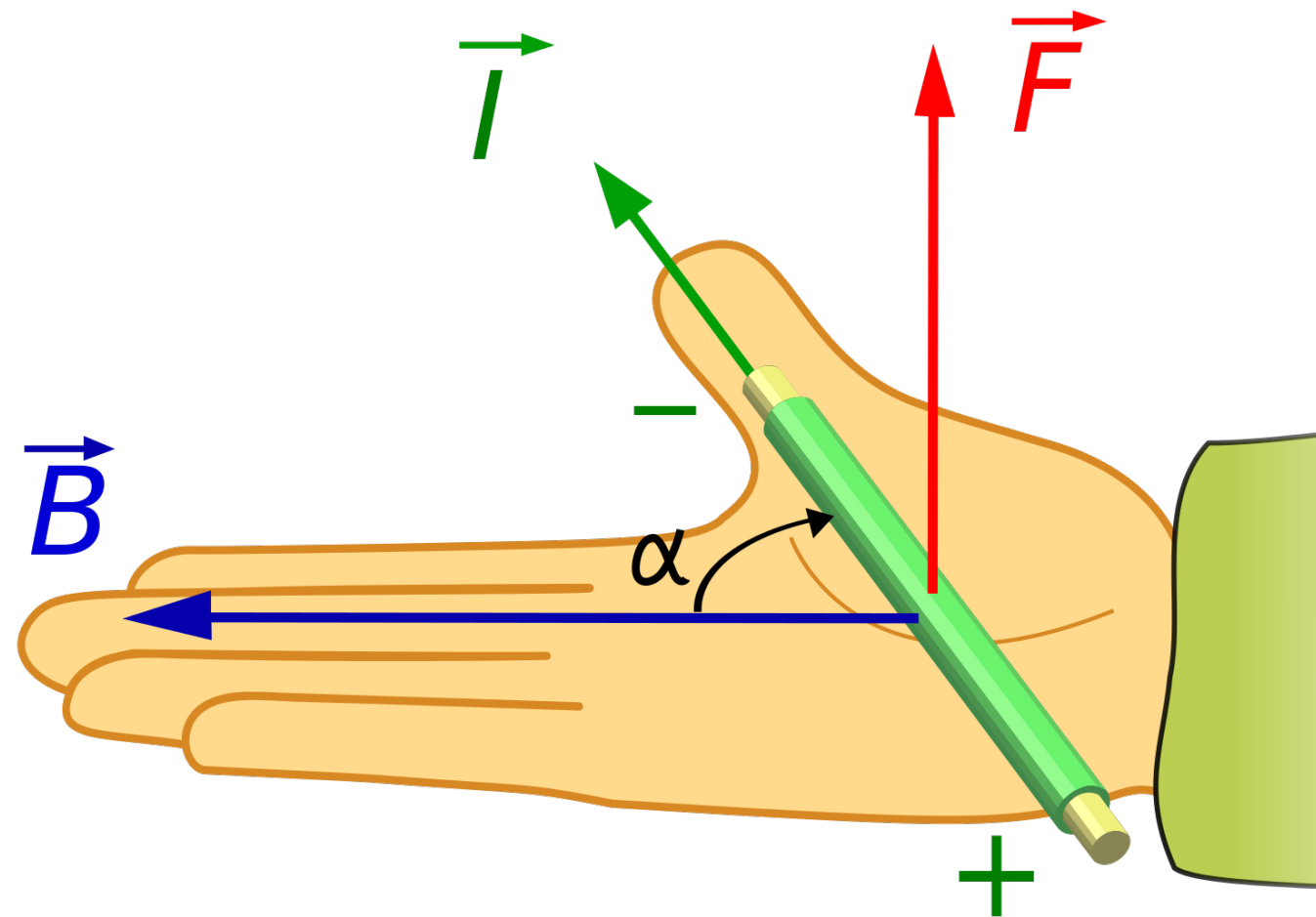
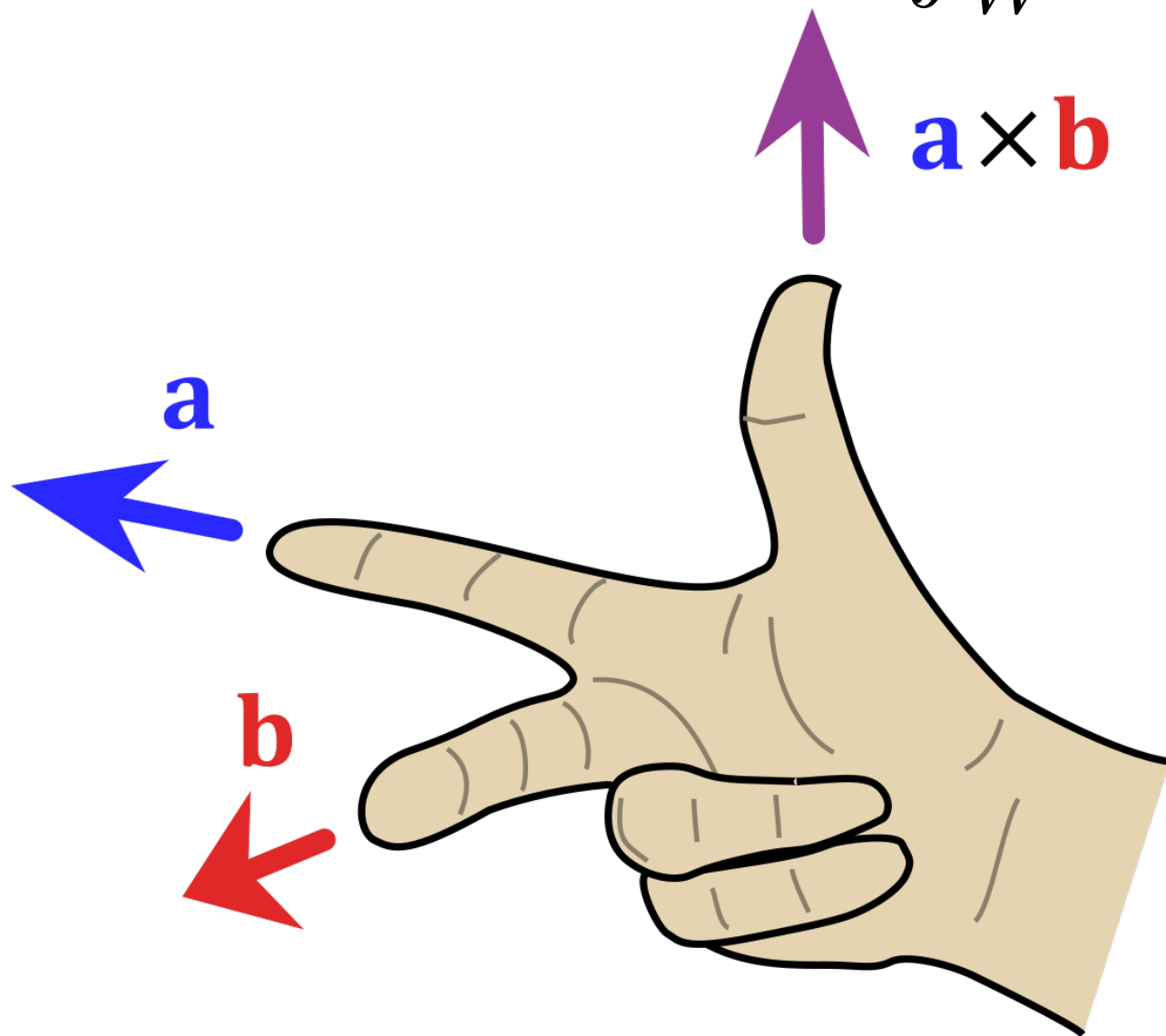
Brush

Commutator

Lorentz Force

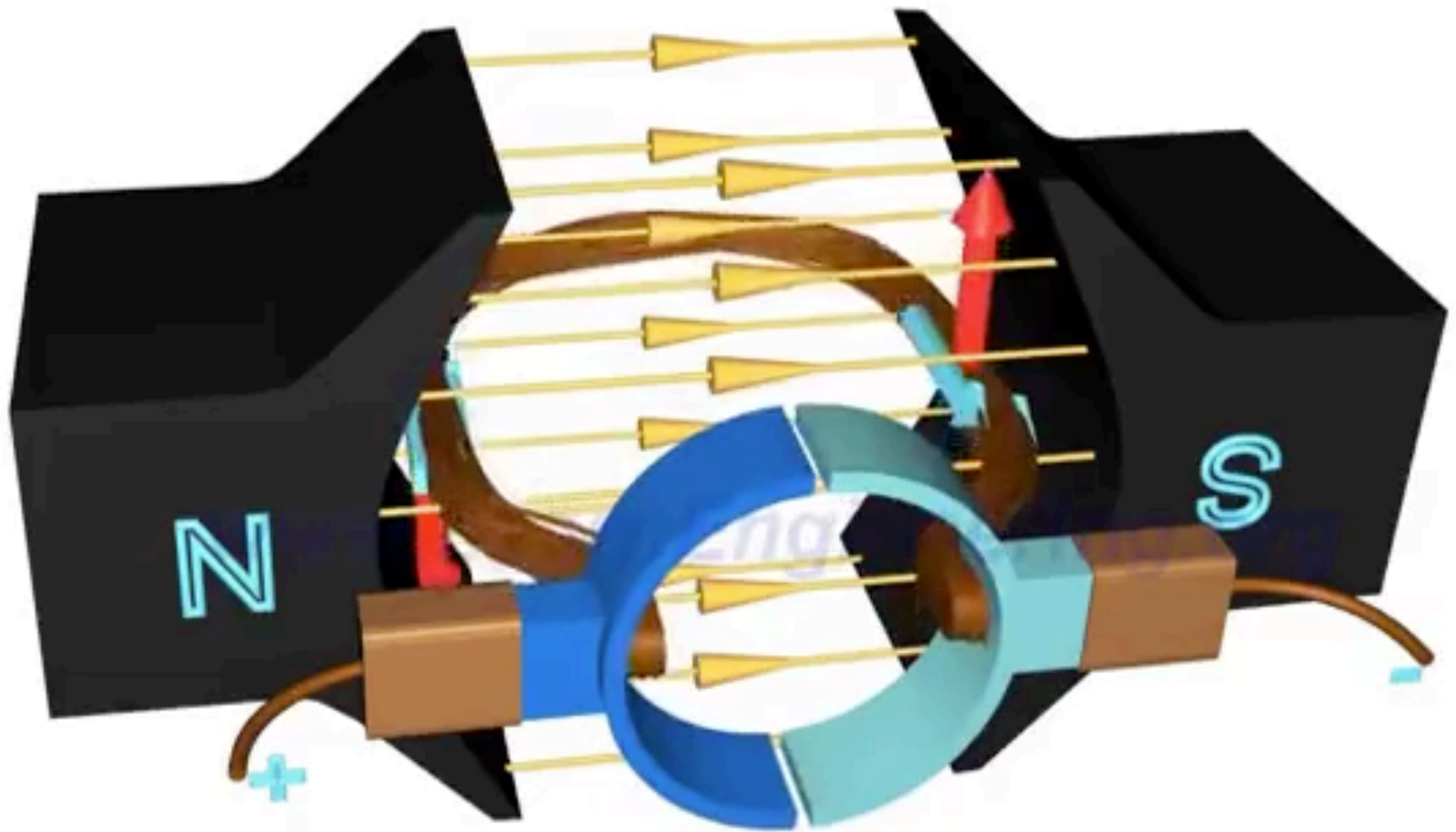
- Current in magnetic field creates a force

$$\mathbf{F} = I \int_W d\boldsymbol{\ell} \times \mathbf{B} = I\boldsymbol{\ell} \times \mathbf{B}$$



DC Motor

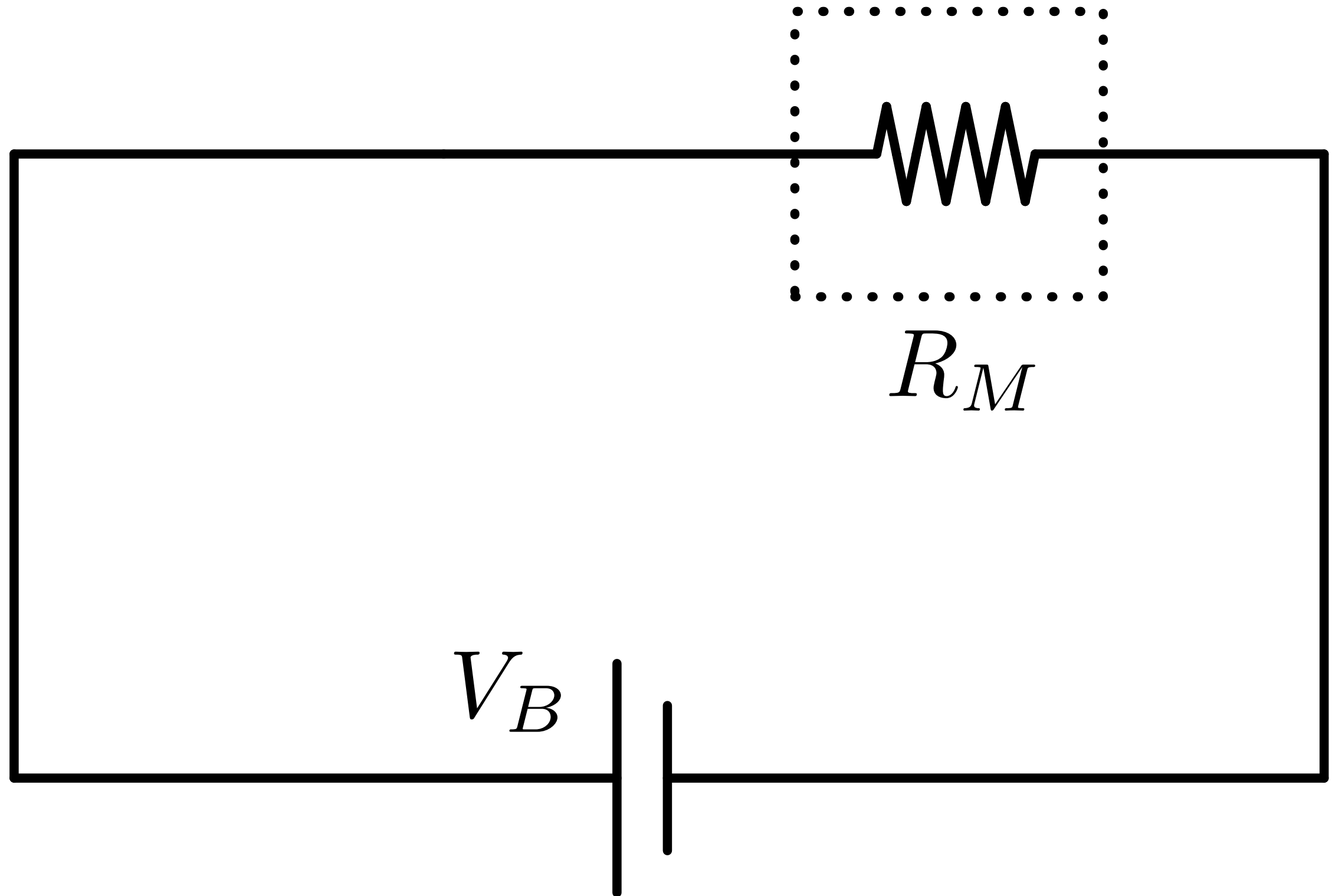
$$\mathbf{F} = I \int_W d\boldsymbol{\ell} \times \mathbf{B} = I\boldsymbol{\ell} \times \mathbf{B}$$



$$\tau \propto I$$

DC Motor

$$\tau \propto I$$

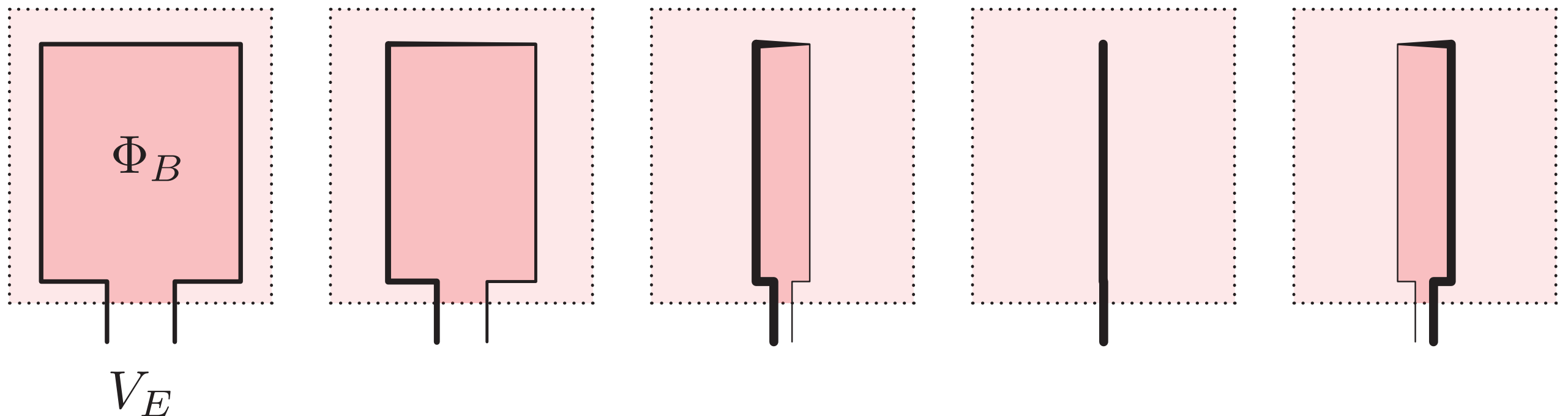


Faraday's Law

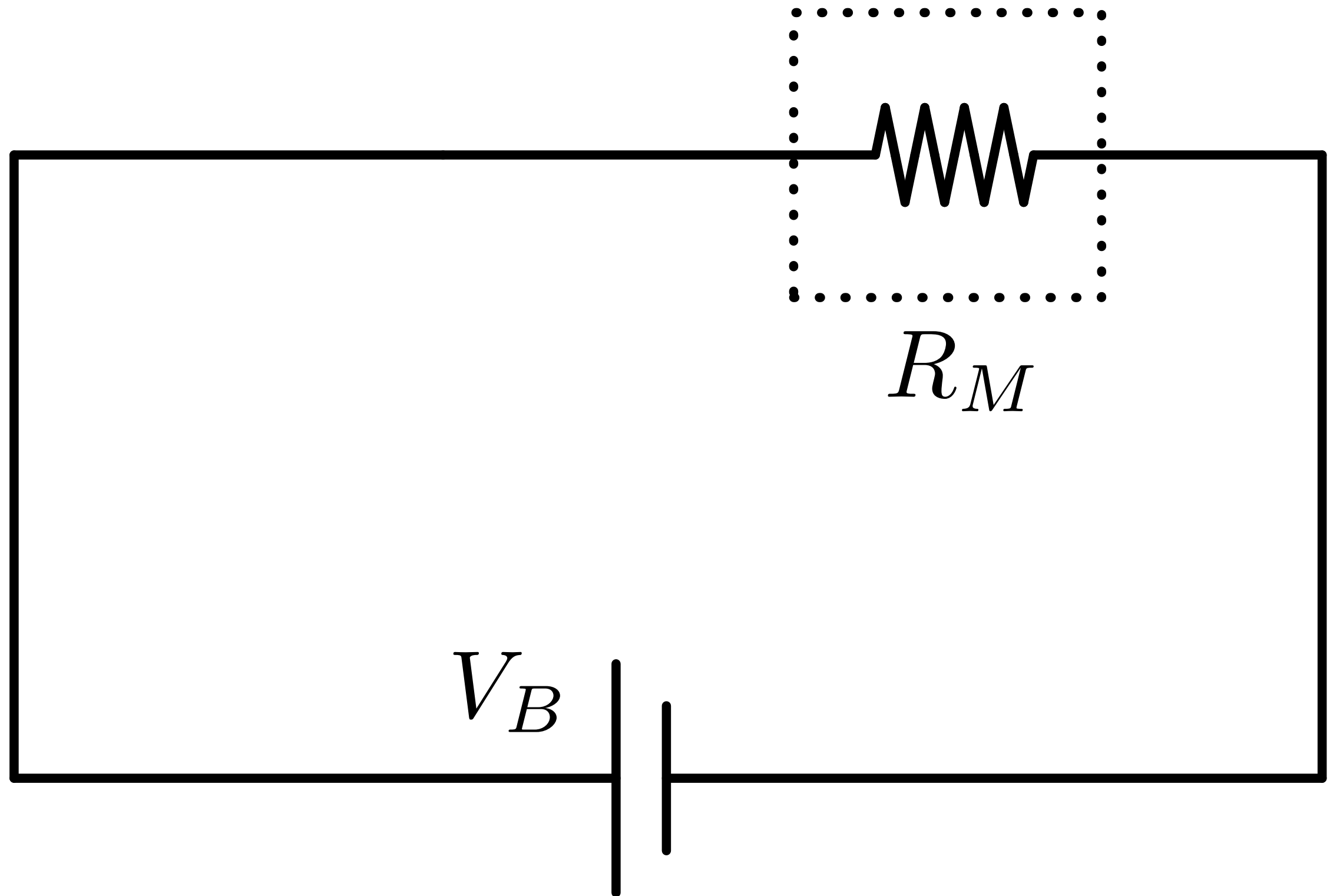
$$\oint_{\partial \Sigma} \mathbf{E} \cdot d\boldsymbol{\ell} = -\frac{d}{dt} \iint_{\Sigma} \mathbf{B} \cdot d\mathbf{A}$$

Back EMF $V_E = -\frac{d}{dt} \Phi_B \propto \omega$

- Change in magnetic flux creates voltage

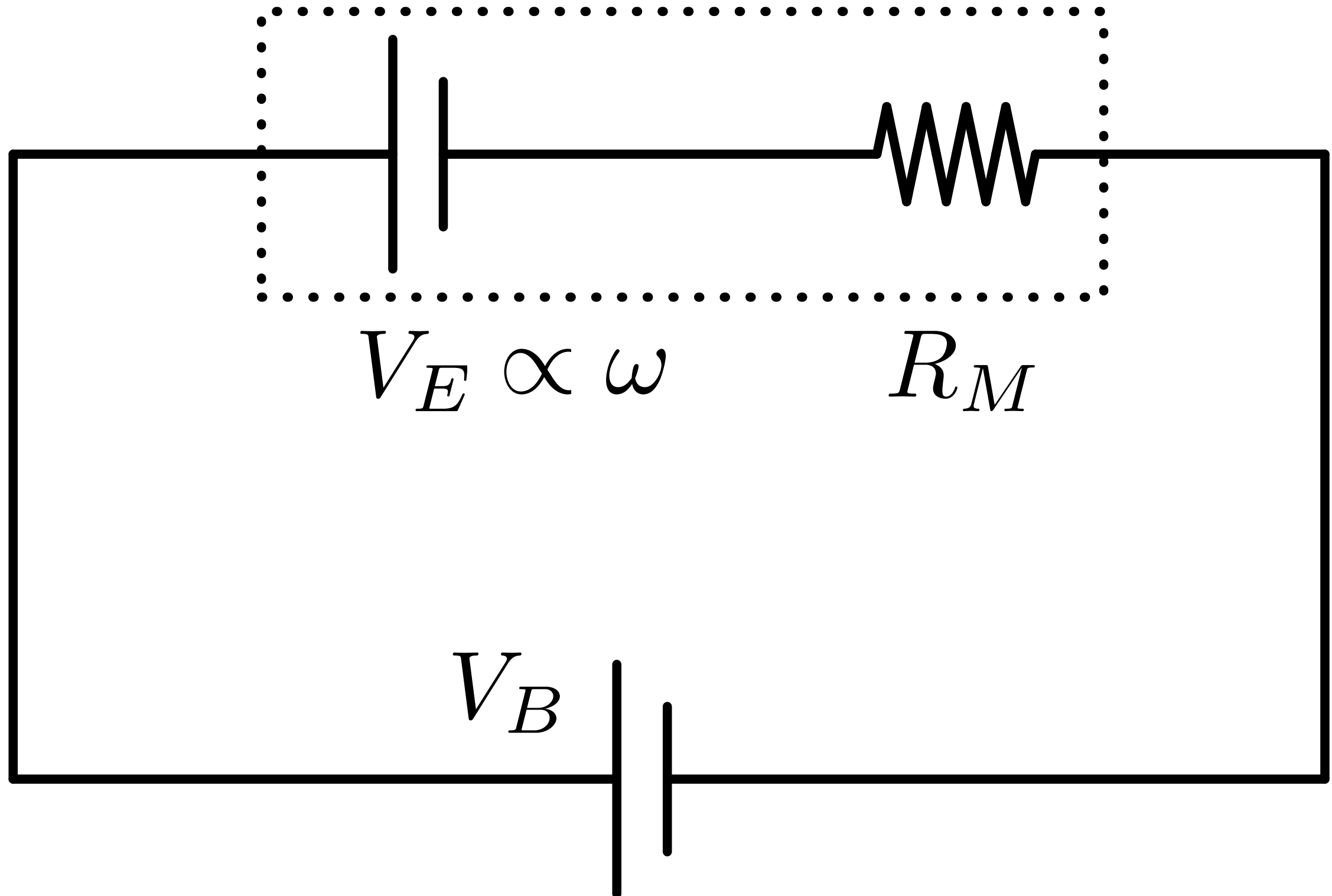


DC Motor

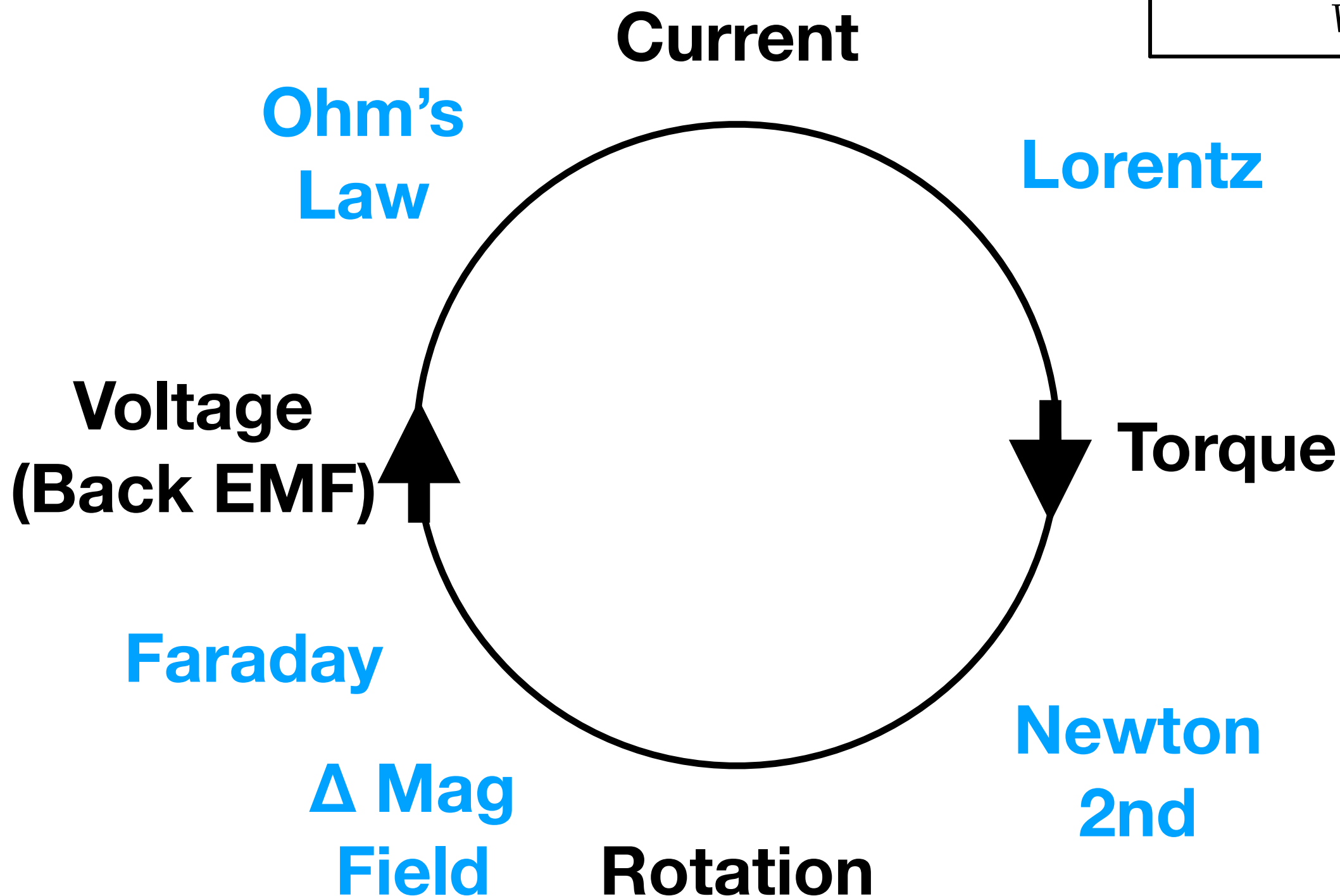
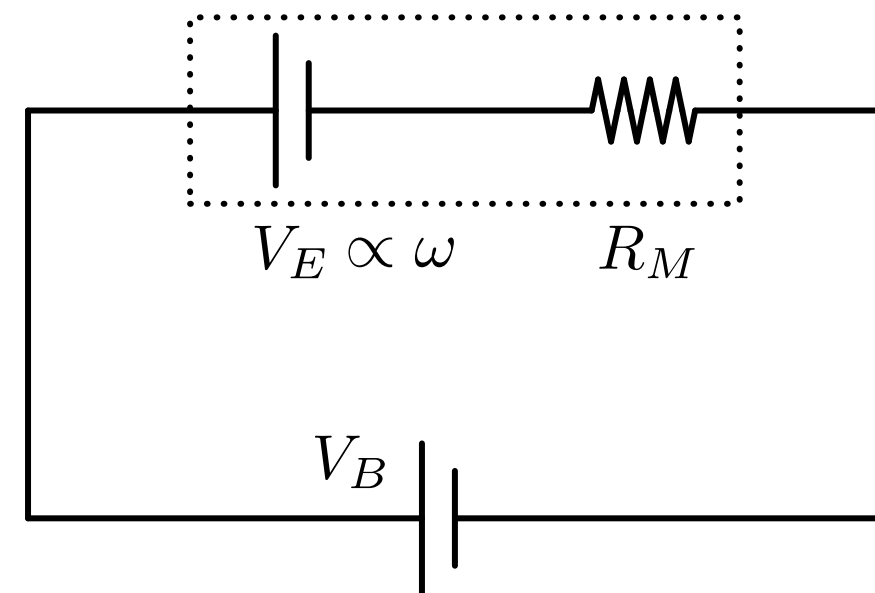


DC Motor

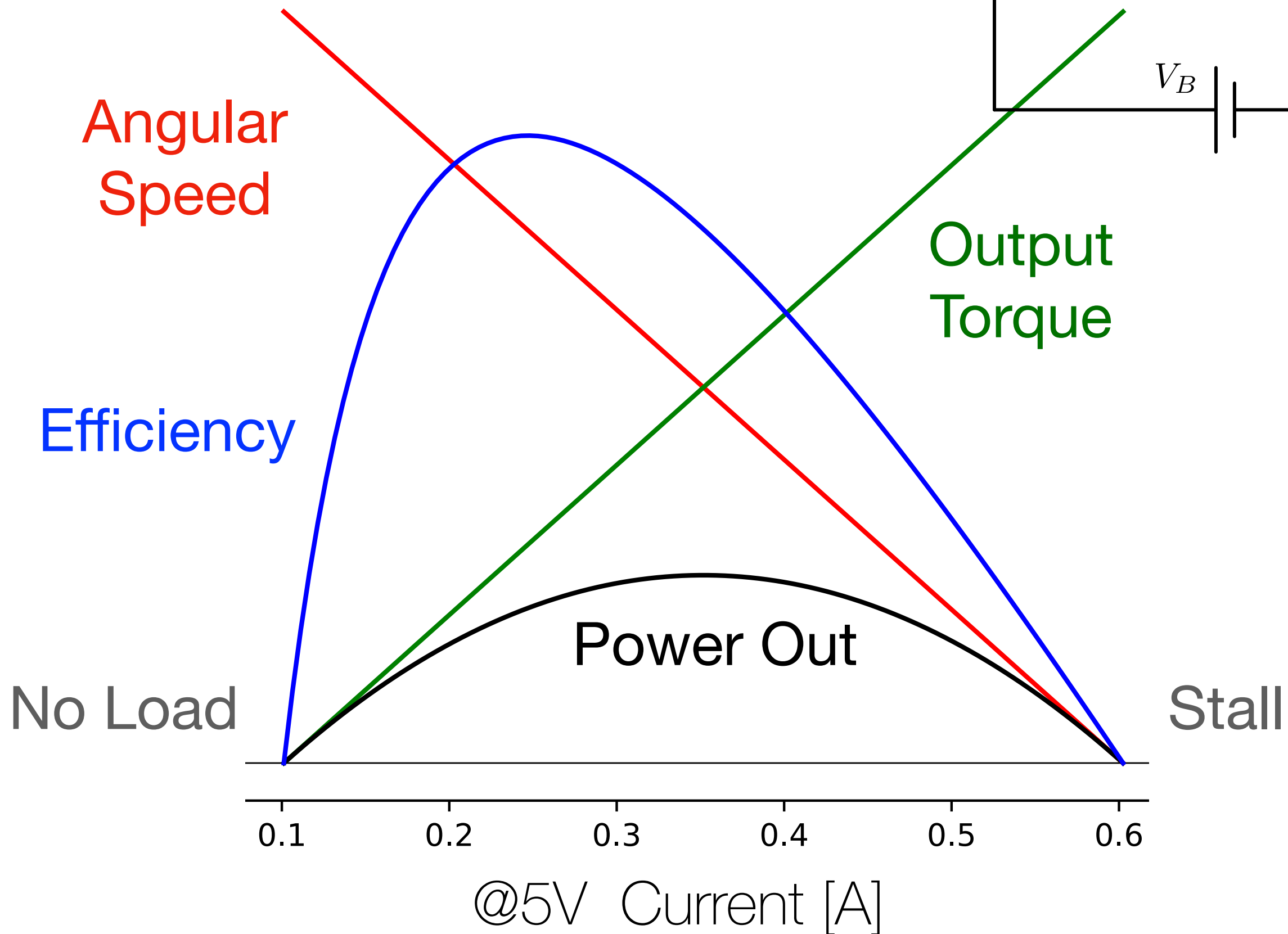
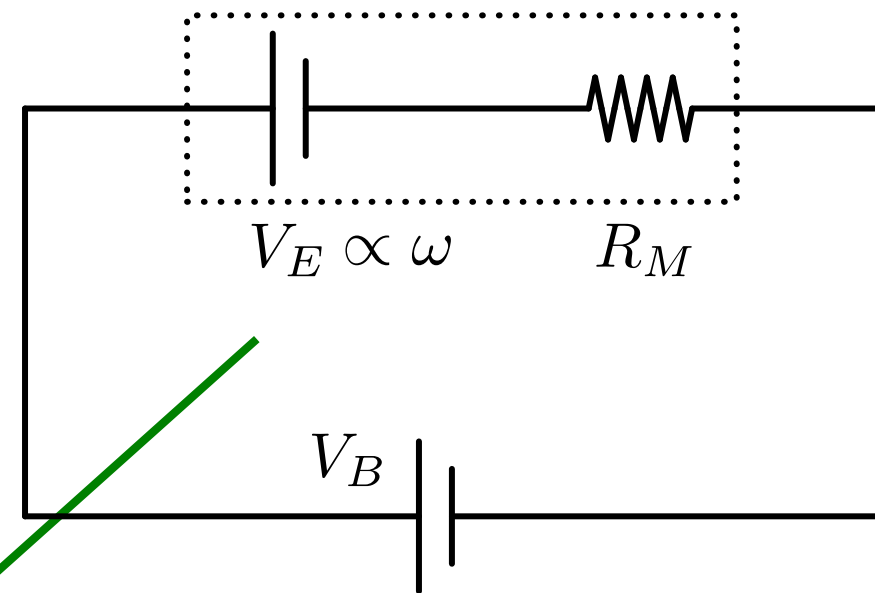
$$\tau \propto I$$



What causes what?

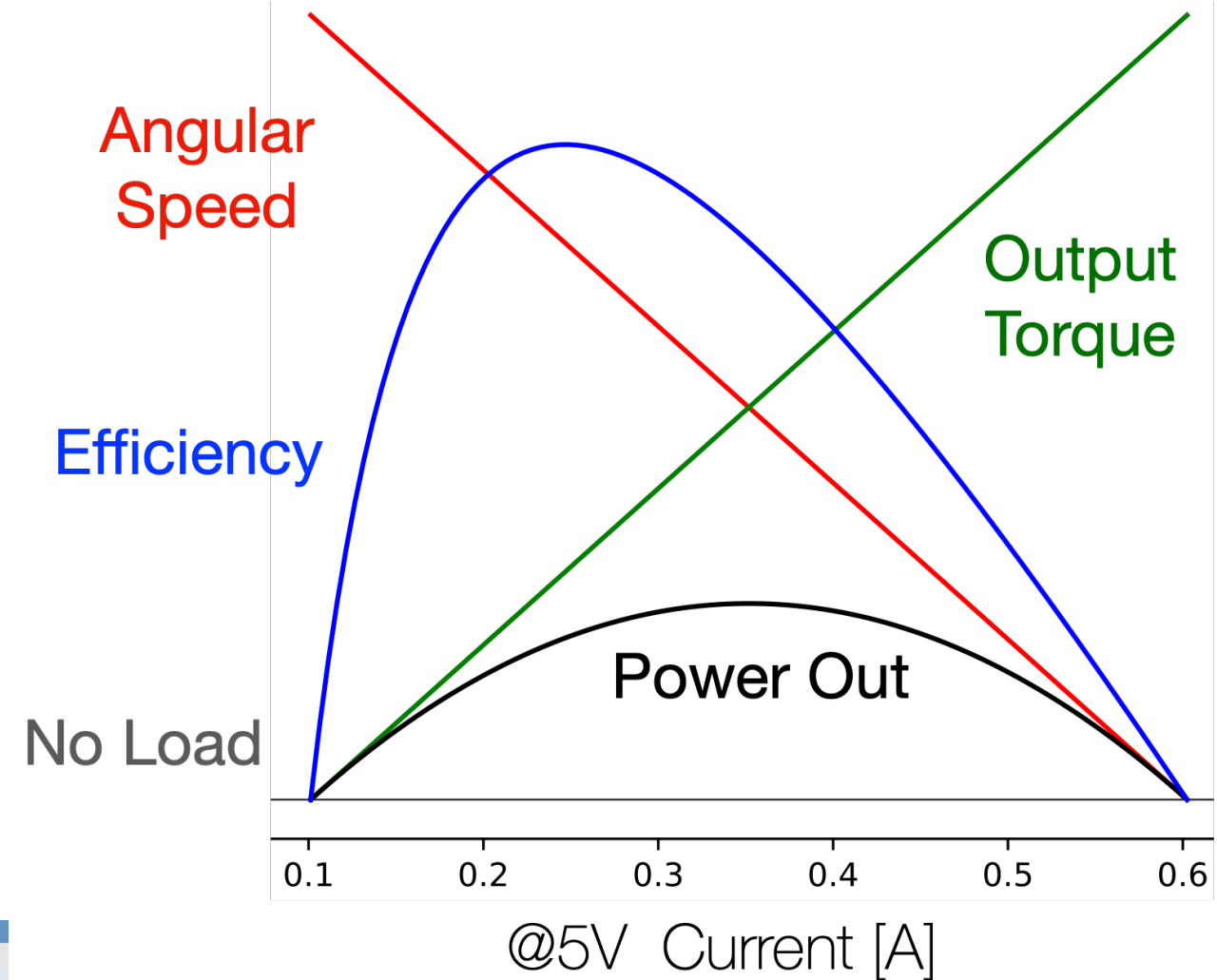


DC Motor



Practice

When a motor stalls,
its current draw is:



A

B

C

D

- A) Maximum
- B) Minimum
- C) Neither maximum nor minimum

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